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arXiv:2407.16780 (q-fin)

[Submitted on 23 Jul 2024]

The Hybrid Forecast of S&P 500 Volatility ensembled from VIX, GARCH and LSTM models

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Predicting the S&P 500 index volatility is crucial for investors and financial analysts as it helps assess market risk and make informed investment decisions. Volatility represents the level of uncertainty or risk related to the size of changes in a security's value, making it an essential indicator for financial planning. This study explores four methods to improve the accuracy of volatility forecasts for the S&P 500: the established GARCH model, known for capturing historical volatility patterns; an LSTM network that utilizes past volatility and log returns; a hybrid LSTM-GARCH model that combines the strengths of both approaches; and an advanced version of the hybrid model that also factors in the VIX index to gauge market sentiment. This analysis is based on a daily dataset that includes S&P 500 and VIX index data, covering the period from January 3, 2000, to December 21, 2023. Through rigorous

testing and comparison, we found that machine learning approaches, particularly the hybrid LSTM models, significantly outperform the traditional GARCH model. Including the VIX index in the hybrid model further enhances its forecasting ability by incorporating real-time market sentiment. The results of this study offer valuable insights for achieving more accurate volatility predictions, enabling better risk management and strategic investment decisions in the volatile environment of the S&P 500.

Comments: 38 pages, 18 figures, 14 tables
Subjects: **Trading and Market Microstructure (q-fin.TR)**; Portfolio Management (q-fin.PM); Statistical Finance (q-fin.ST); Machine Learning (stat.ML)
Cite as: **arXiv:2407.16780 [q-fin.TR]**
(or **arXiv:2407.16780v1 [q-fin.TR]** for this version)
https://doi.org/10.48550/arXiv.2407.16780
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Submission history

From: Robert Ślepaczuk Ph.D. [view email]
[v1] Tue, 23 Jul 2024 18:28:16 UTC (1,753 KB)

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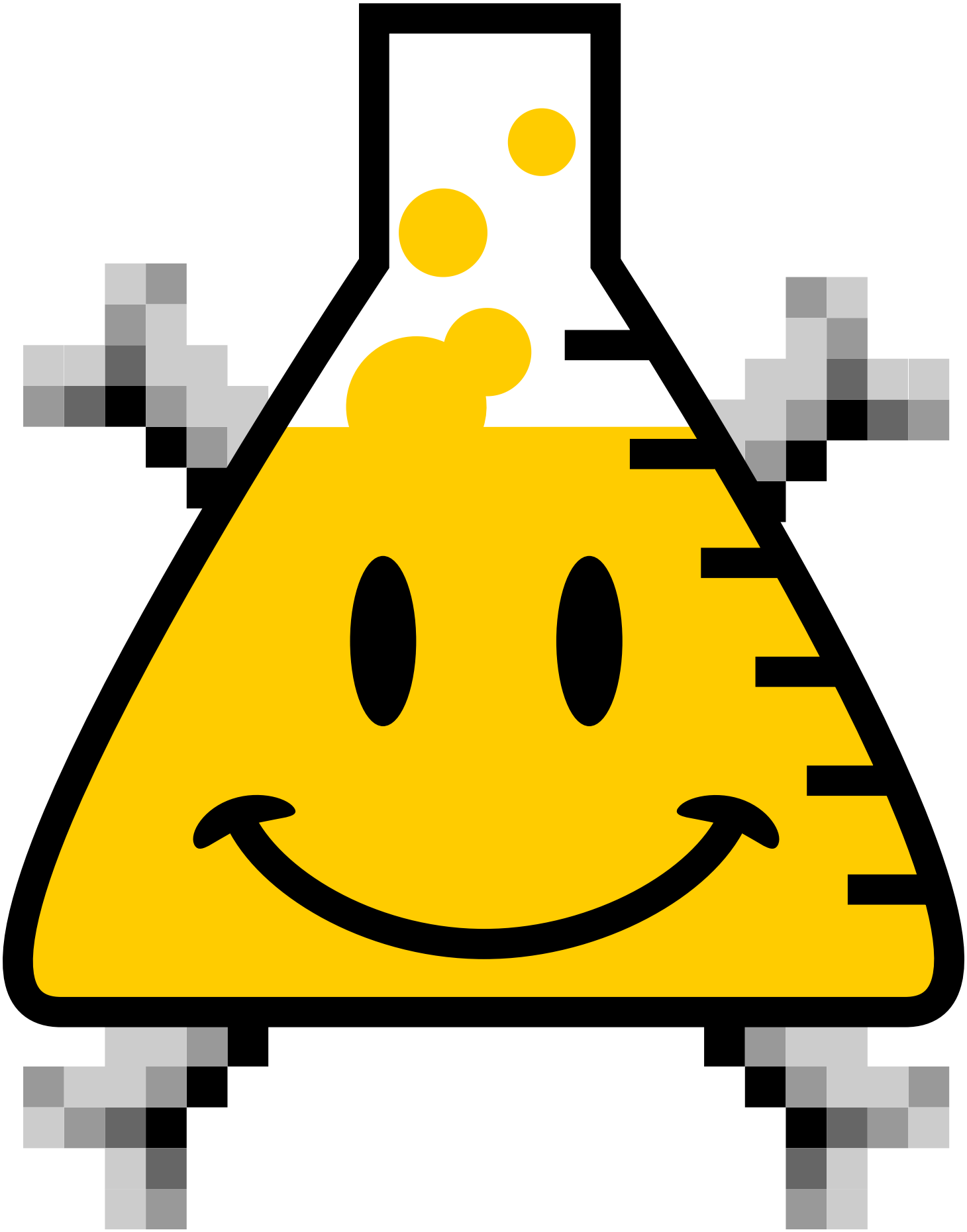

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